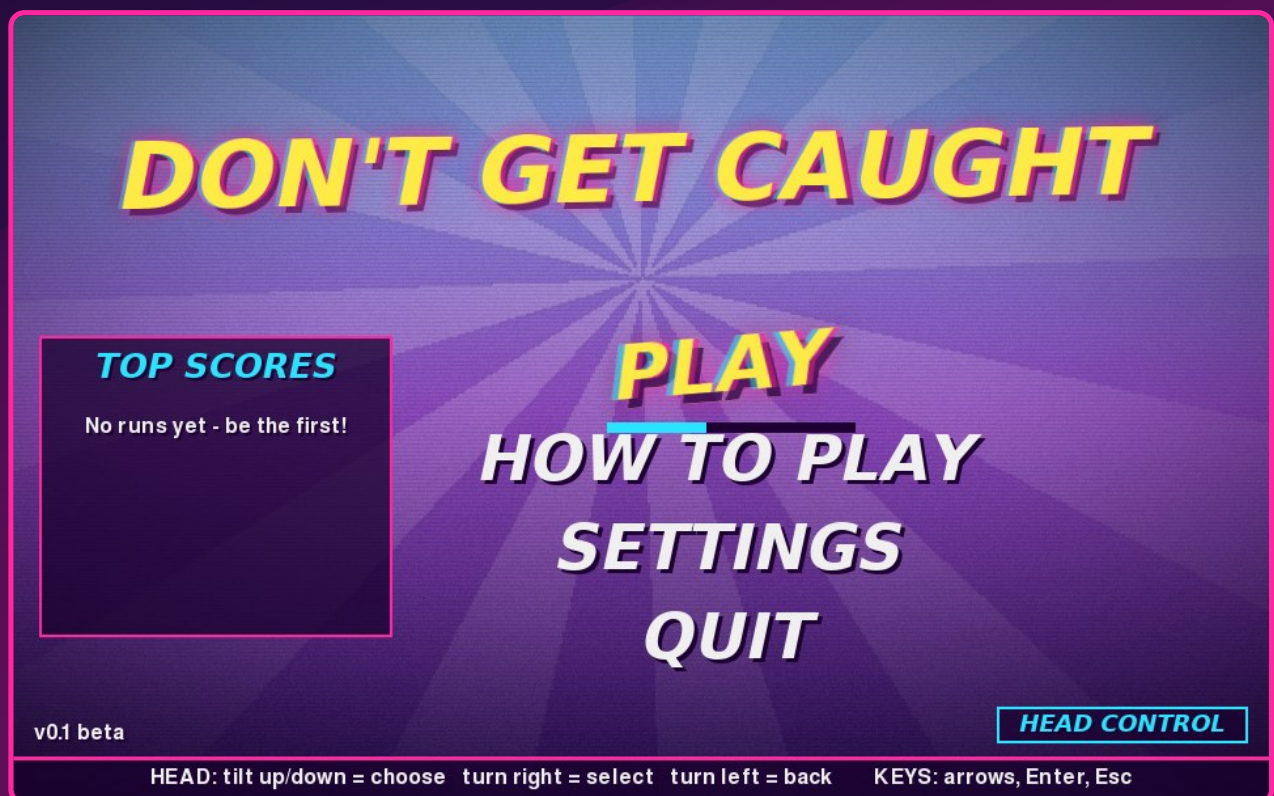


DON'T GET CAUGHT

Development Report, Commit by Commit

The design, the architecture and the story of 64 changes



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01

Introduction

In 10 days, from 30 September to 9 October 2026, "Don't Get Caught" grew from a skeleton-tracking demo into a full game: three exams, characters, music, played by moving your head. This report follows that path through **64 change entries**, one by one.

In this project the design matters more than the architecture, so the report first describes the artistic design, then today's architecture, then gives a short summary and lessons learned; the 64 entries come last, one by one, in order.

Project: the ME461 course, team **Glitch Please**. The player sits an exam in front of a webcam. Turning the head left or right copies from a neighbour's paper, looking down writes on your own paper, looking at the screen watches the teacher. Copying while the teacher looks, or staring at him too long, gets you caught.

How to read it:

- The numbers are the "Commit #N" entries in `NOTES.md`. Git has 21 commits; one git commit usually holds several entries. Each section says which entries went into which git commit.
- Each entry is summed up from three angles: **Technical** (what was coded, which numbers), **Architecture** (how the files and modules changed), **Art** (pictures, sound, music, feel). An angle with nothing to say is left out.
- #58–#64 are not committed yet; they are in the working copy.
- The screenshots were taken from the game's own drawing code (without a camera, with sample data).

Technology: Python 3.14, `opencv-contrib-python` (camera), `mediapipe` 1.0 Tasks API (face points and head rotation), `pygame-ce` (window, drawing, sound), `numpy`. 315 unit tests run with the code.

02

Artistic design

The game's visual language is made of three deliberate worlds, each serving a different feeling: the **illustrated classroom carries tension**, the **code-drawn neon carries excitement**, and **paper and ink carry officialdom and defeat**. Moving between them, the player can tell from colour and texture alone what to feel.

Above all of them sits a **retro arcade theme**: not a generic, modern arcade, but an 80s arcade hall: old-TV scanlines, neon colours, synth sounds, a top-5 score table with three-letter names entered like on an arcade cabinet, points counted up with chip-like ticks, slot machines and "JACKPOT!". The game turns an exam into a machine in that hall.

■ 1. The illustrated classroom: tension

The heart of the game is the classroom seen first-person from the player's seat. The team made the pictures with Gemini, in a comic-book style: dark outlines, soft flat shading, warm and muted colours, blue-grey walls, green chairs.

- **The teacher is one consistent character:** long grey hair, a grey beard, a tightly tied camouflage bandana, a black T-shirt, jeans. Every prompt repeats this description so he is the same man in every scene.
- **The picture set:** the classroom in four states (board/desk × busy/watching), three look-away pictures (your paper, the left and right neighbours), 10 neighbour papers (A–D and "?"), 8 Greek-letter variants for the not-a-ME-student character (α β γ δ), the warning and tearing scenes, the mugshot papers.
- **A picture is a contract with the code:** the code looks at fixed spots in the pictures (the warning scene zooms into the teacher's face, the neighbours' papers sit at fixed places, the mugshot photo lands on a pure green area). So every prompt in `IMAGE_PROMPTS.md` starts with "keep the camera angle, scale and the teacher's head in place", and every result is checked against its source.
- **The unseen teacher:** the classroom is only visible while the player looks at the screen; while copying, the teacher can only be heard. The visual design builds tension by hiding information on purpose.



The warning scene's picture: the teacher at your desk, pointing at you (classroom_warning)

■ 2. Code-drawn neon: excitement

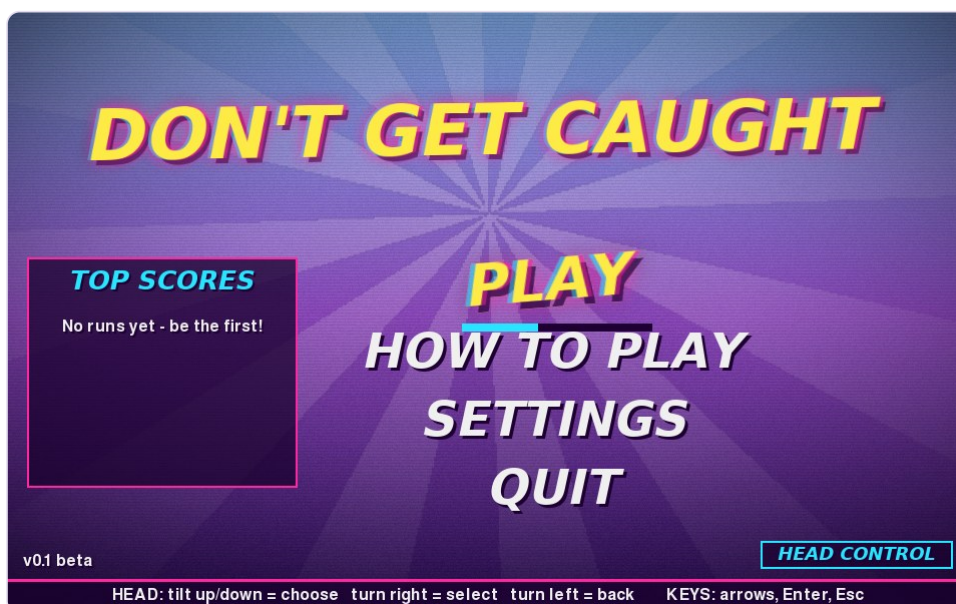
The menus, the briefing, the slot machines, the character portraits and the Gemini/Claude logos live in an 80s world inspired by Hotline Miami, and are **drawn entirely in code**; this world has no picture files.



Inspiration: the main menu of Hotline Miami 2: Wrong Number (Dennaton Games / Devolver Digital). The sunset-coloured background, slanted flickering neon text and VHS texture are where our menus started.

- **Palette:** five neons on dark purple: pink (255, 40, 160), cyan (40, 230, 255), yellow (255, 235, 70), green (80, 255, 150), red (255, 50, 80). Green means good, red bad, yellow emphasis (the gossip's +/- lines, the score table).

- **Typography:** titles in Impact (or Arial Black), slanted like Hotline Miami. Behind every text are half see-through pink and cyan "ghost" copies shifted 4 pixels and a blurred pink glow. The official notice is different on purpose: a typewriter font (Courier).
- **Background layers:** a three-colour sunset gradient drawn one pixel row at a time, soft-edged light rays, a light film grain that hides colour steps, an old-TV scanline every 3 pixels, darker corners.
- **Elements:** the MOOD-O-MATIC slot machine (dome, gold-framed reel, glass gleam, lever, casino bulbs every 24 pixels along its edge); spinning per-digit reels for the score and "JACKPOT!"; neon portraits of the ten characters (New Era cap, glasses, Monster laptop bag, a "#7" mug...); Gemini's many-coloured four-pointed star (a superellipse) and Claude's warm rays, both with cartoon faces.



Main menu: sunset, light rays, slanted neon text



Character choice: portraits drawn in code

■ Score: inspired by Balatro

As the game's author, I really like Balatro and its scoring: the points do not appear at once, they are counted part by part and climb a little at every step, which gives the player a small victory every time. The score screen and the bonuses were designed the same way.



Inspiration: scoring in Balatro (LocalThunk / Playstack, Steam store screenshot). Chips and mult (340 × 21,600) grow step by step with every card; our score table and slot reels aim for that feeling.



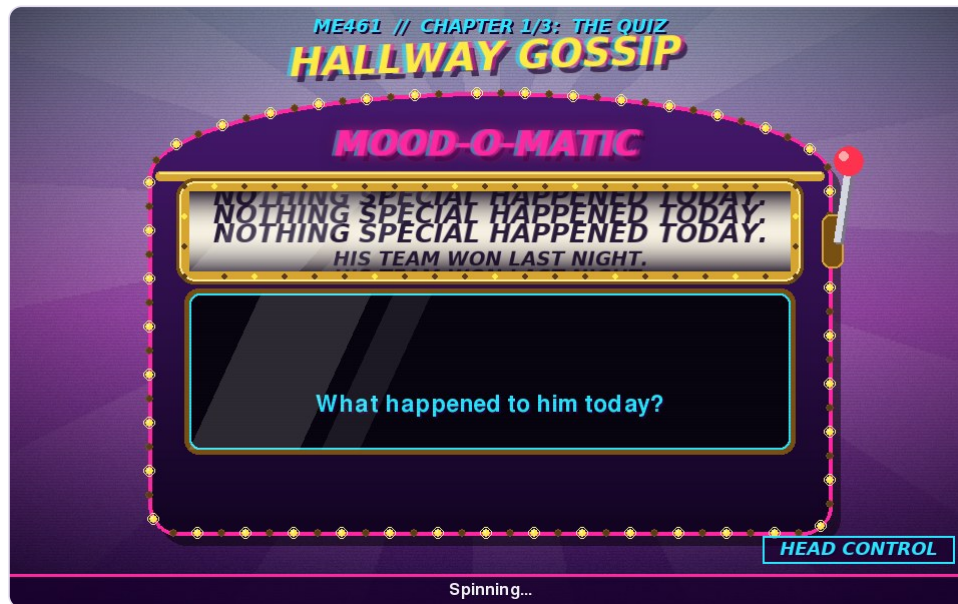
Our score count, mid-count: the table lights up row by row, the points pop in, the slot reels below spin.



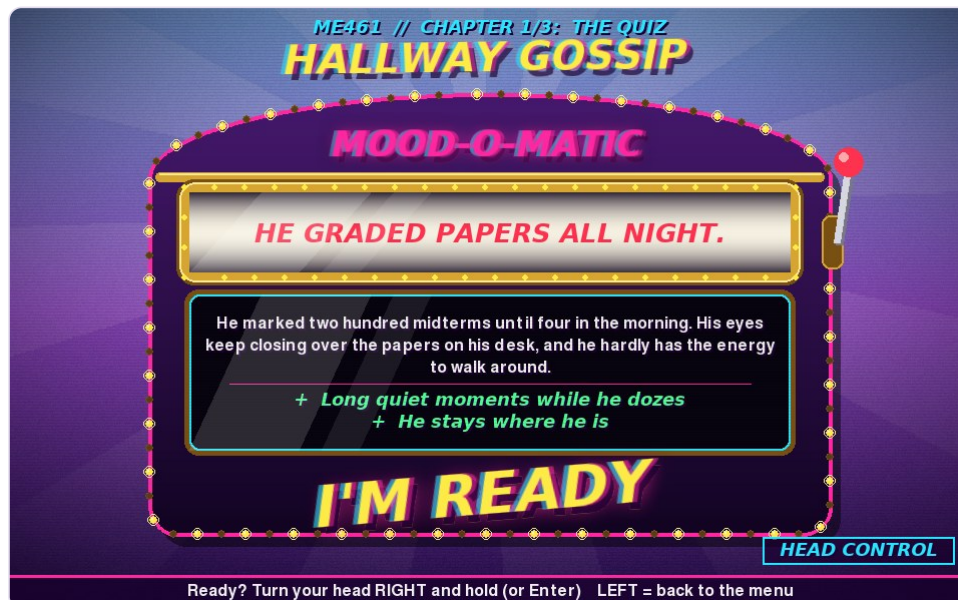
The same screen when the count is over: the reels have landed on the score, with the run total and the class average under it.

- **Counted part by part:** after an exam the score is counted row by row in a table; the row whose turn it is glows in its colour, its points pop in and join the big score, and each tick is a semitone higher.
- **Balatro-like bonuses:** **NINJA!** +1500 (every answer right, no warning), **ALMOST NINJA** +500 (one warning), **SHARP EYE** +100 (finding the neighbour who knows at the first look), **RAZOR CLOSE!** (getting away with the suspicion bar above 80 %, with a heartbeat), the early hand-in bonus; and penalties: warnings, the nerd's "NERD WAS LATE" -1000.
- **The bigger the score, the bigger the show:** the big number is a row of slot reels, one per digit; as the score climbs the reels shake harder, the bulbs chase faster, more sparks fly, and a huge score ends with "JACKPOT!". The semester grade follows the same idea: from BB to AA the confetti, light rays, explosion and fireworks grow step by step.

Why slot machines? Apart from gambling, a slot machine is an aesthetically very attractive object: the bulbs, the spinning reels, pulling the lever, the reel clunking into place and the jackpot moment are a language of excitement everyone knows. That language also fits a high score very well: reels spinning and stopping one by one while the points are counted make a big number feel like a win. There is no money or betting in the game; the machine is only there to draw the teacher's mood (the MOOD-O-MATIC) and to show the score.



The MOOD-O-MATIC spinning: the moods on the reel blur past, the casino bulbs chase round its edge.



The MOOD-O-MATIC stopped: today's mood in red in the middle, the story and the +/- lines on the screen below, "I'M READY" under it.



The AA show: an explosion, turning gold light rays, fireworks, confetti and "LEGENDARY CHEATER".

■ The talking AIs: Gemini and Claude

The two AIs that made the game are characters in it: Gemini (the pictures) and Claude (the code). Both are logos drawn in code with cartoon faces; Gemini talks in blue speech bubbles, Claude in orange ones, and every letter typed makes a small talking blip in that speaker's own pitch.

They were first conceived as a "How to play" mechanic. Explaining the rules as a list of text was dull, and in a game played with the head, reading is the weakest way to learn. Instead, two characters explain the game in a more fun mood, through a lively conversation:

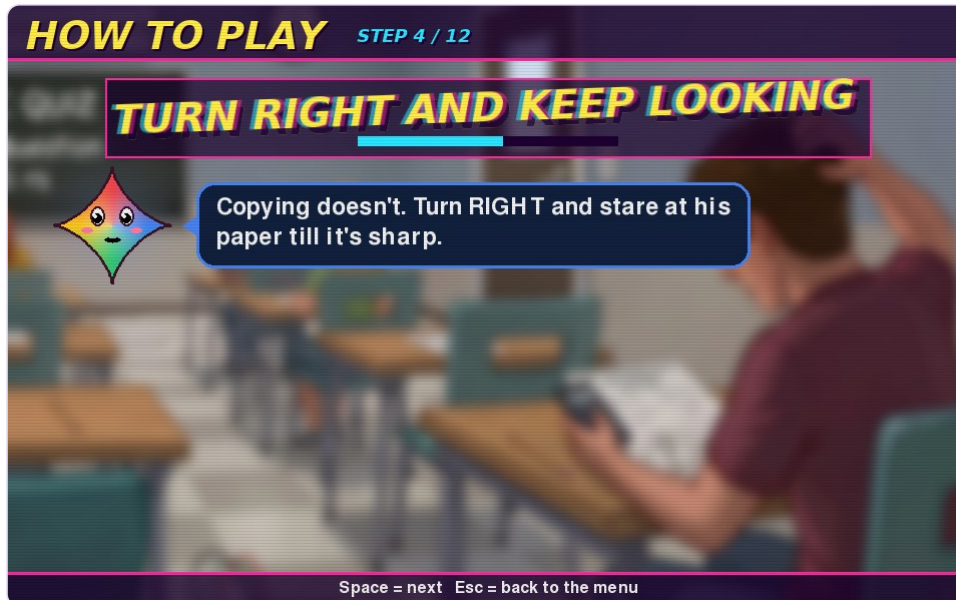
- **They take turns, finish each other's sentences and tease each other.** The guide flows less like a lesson and more like a scene where two people help the player while making fun of them.
- **The player tries everything with their head as it is explained.** Behind them is what the player would see in the game: looking down shows the paper and what they wrote, turning right shows the neighbour with "?" getting sharper, turning left the one with "B", looking at the screen the teacher, busy and then turning with the "hmm". A banner says the task, a bar fills while the head holds the right direction, and "NICE!" when it is done.
- **Short and ironic.** The first version was 41 lines, about 2 minutes; it was found too long and cut to 23 lines, about 45 seconds. It still teaches every mechanic but gives its tips as jokes: "guessing loses points: it's called math", "copy NOW, not 'in a sec'", and for the hidden

suspicion line, "I put a random line in the code." "Where is it?" "No idea. I forgot. Good luck!".

- **The guide ends in a real exam:** a two-question practice exam that counts for nothing, so what was learned is used at once.



The guide, first step: Gemini and Claude introduce themselves over the game's classroom.



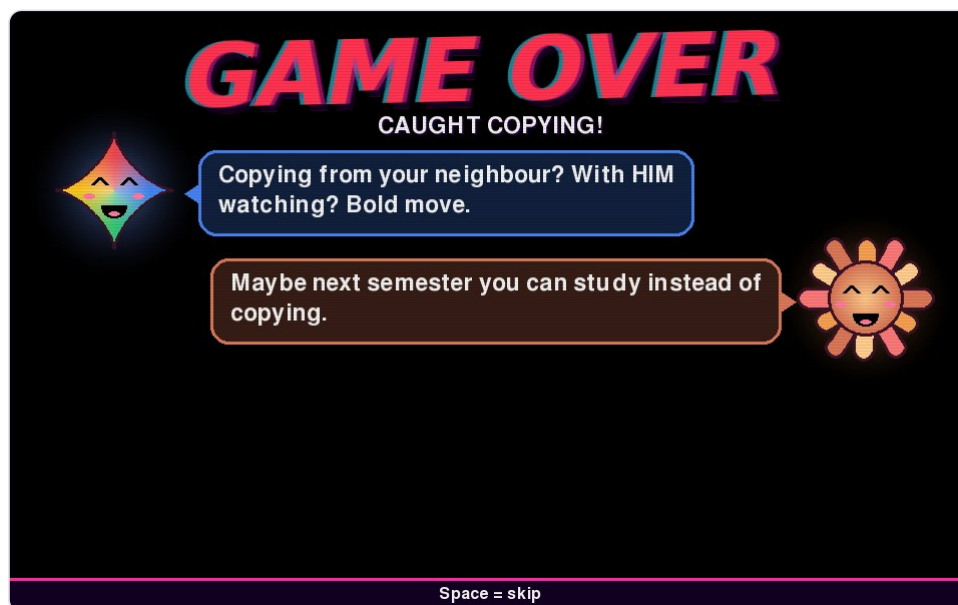
The guide, a task step: the task banner and its filling bar on top, the right neighbour's paper getting sharper behind.

Then they moved to the end of the game too. Since they worked so well in the guide, we thought the same duo would be a lot of fun at the end as well:

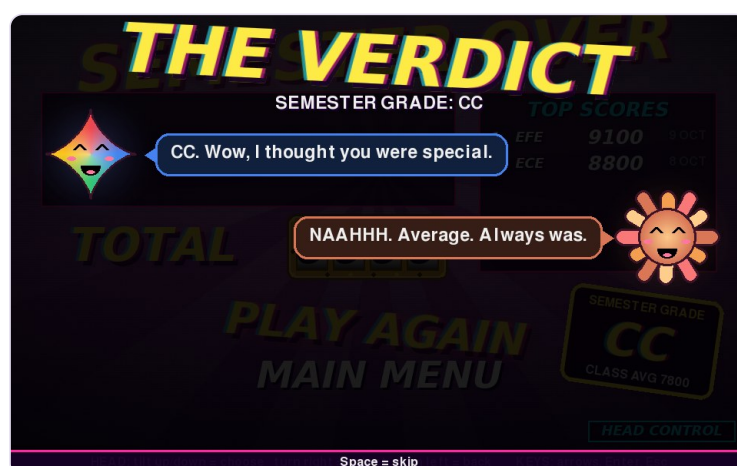
- **The game over chat:** after a loss, on black, the two make fun of you depending on how you lost (separate conversations for being caught and for warnings, 8 of each). They are drawn

from a "bag": every joke comes once before any repeats, never the same twice in a row. After the second line both laugh together.

- **The grade roast after the Final:** once the semester grade is stamped, the two come back and roast the letter in two or three ironic lines ("CC. Wow, I thought you were special." / "NAAHHH. Average. Always was."; "AA? ... You are... 'the one' :O"). A good grade gets a growing show first; it can be turned off in Settings.
- **Why AIs?** The team wanted the two AIs that made the game to laugh at the cheater. It is a meta joke: the makers of the game judge the player. And bringing back at the end the characters met in the guide gives the game the feel of a story with a beginning and an end.



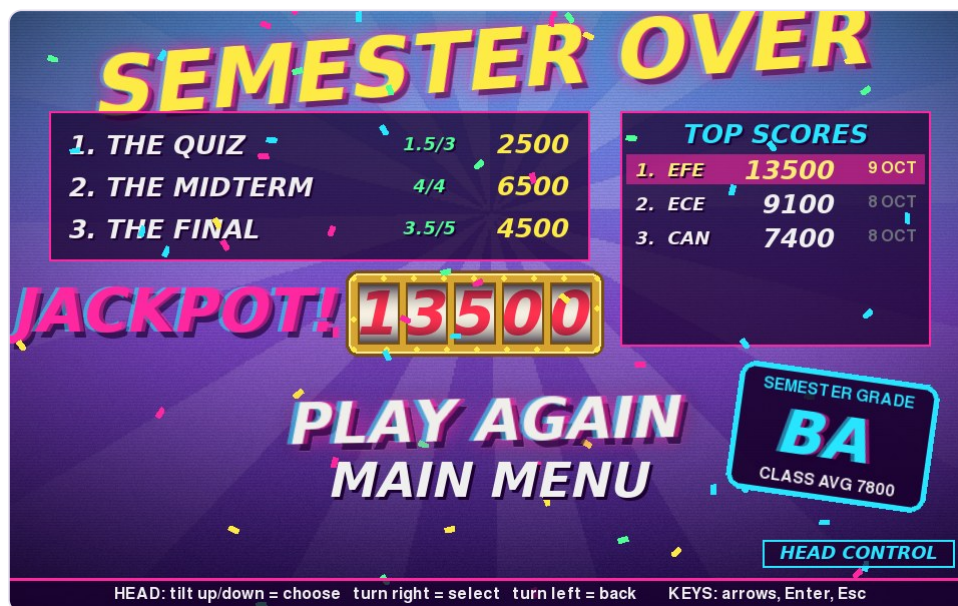
Game over chat: after being caught, Gemini and Claude make fun of you.



The grade roast after the Final, for a CC: "CC. Wow, I thought you were special." / "NAAHHH. Average. Always was."

■ Competition: the curve and arcade names

- **Grades on a curve.** The run's letter grade (AA...FF) does not come from a fixed table but from all earlier runs played on this computer: how many standard deviations the total is above their average; an average run is a CC. So the grade depends not on how many points you got but on **where you stand against the other players**. This strengthens the competition between friends playing on the same computer: the better one plays, the harder an AA gets for everyone, just like in a real class. For the first 5 runs there is no "class" yet, so a fixed table is used.
- **Three-letter score entries.** A player who makes the top 5 enters a three-letter name like on old arcade cabinets: tilting the head up and down rolls the letter, turning right goes to the next one. It is the most direct piece of the retro arcade theme and gives the score table the feel of an arcade hall's wall ("1. EFE 12808 7 OCT").
- **The secret reason:** the real, hidden reason for three letters is that the game's author's name has three letters too: **EFE**.



The end of the semester: the three exams' results, "JACKPOT!" on the slot reels, the top scores with three-letter names, and a BA stamped on the curve against the class average.

■ 3. Paper and ink: officialdom and defeat

The opening and the moment of losing are cut off from the neon world on purpose.

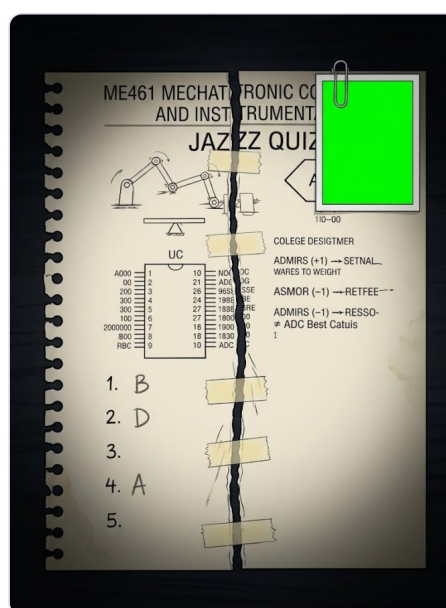
- **The official notice:** a cream paper slides onto a wooden desk drawn line by line in code; "ACADEMIC INTEGRITY DEPARTMENT", lines typed out on a typewriter, a shaky signature

that is someone trying to draw a helicopter, and a red "APPROVED" stamp slamming down with a shake. The satire starts here: the game is fiction, and so is the teacher ("Probably.").

- **The mugshot:** when you lose, the screen goes dark and silent; the game's JAZZ QUIZ paper, torn in half and taped back together, fades in under a single interrogation lamp. Your face from the moment you lost is clipped to it; a moment later the teacher writes "GOT CAUGHT!" over it in red marker. The opposite of the neon: no colour, no music, only the squeak of the marker. I put the mugshot in mainly because it is funny and fun: the player's surprised face from the moment they were caught appears on the paper, which turns losing from something annoying into something to laugh at.



The official notice



The mugshot paper: a dark room, one lamp

■ Motion and rhythm

- **Timed to the music:** the briefing and the character screen are locked to the 147 BPM track. Lines slam in on each bar's hit and thump on the beats in between; they start soft, get harder every line and peak with "DON'T GET CAUGHT.". The menu title also thumps 6 % bigger every 0.5 s.
- **The language of impact:** text slamming down from big, a white flash, a screen shake that calms down. The flash also hides cuts between pictures (in the warning scene, from the zoom to the teacher at your desk).
- **Transitions:** every screen change is a 0.4 s crossfade; music fades over 1.5 s. The mugshot is slow on purpose: the paper takes 1.5 s to come out of the dark.

Big effects, on purpose. Screen shakes, flashes, huge slanted text, explosions and confetti are deliberately overdone. I think big, generous effects make the game much more attractive: even a small success becomes a big moment on screen, and the player wants another round. The excess also suits the satire: copying in an exam is staged like a film scene.

■ Humour and tone

The game does not encourage cheating; it satirises it. The tone is the same everywhere: dry, self-mocking, a little desperate.

- **Lines:** "0 HOURS OF STUDYING.", "NO PRESSURE. (A LOT OF PRESSURE.)", "THE FINAL — God, please help me.", "LEGENDARY CHEATER" in the grade show.
- **Gemini and Claude:** the two AIs that made the game mock the cheater at game over and at the semester grade, and finish each other's sentences in the guide.
- **The characters are types from a real class:** the New Era guy, glasses, the nerd, the 7th-year legend, the "quick question" guy... None is named with an insult or could point at real classmates; some ideas were dropped on purpose for that.

■ Sound design

Sound here is gameplay information, not decoration: while copying, the player cannot see the teacher, only hear him.

- **Warning sounds:** Luigi's "hmm" says the teacher is about to turn; the chalk loops while he erases the board and stops the moment he starts to turn, so **silence is a warning too**. While looking down you hear nothing.
- **Sounds made in code:** everything without a file is generated with numpy: 80s synth button sounds, the typewriter, the signature, the stamp, a two-pull tear, footsteps, a heartbeat that pushes the music far away, the logos' talking blips, explosions and fireworks, the whiteboard marker's squeak. Without a sound device or a file the game carries on silently or with a beep.

Music. All three tracks were made with Suno, each by sampling a chosen song:

- **Menu theme** (`theme.mp3`): made by sampling "Bastır" by Murda and EGE! (in an 80s Miami mood). Plays in the menus and between exams.
- **Character music** (`character_[cut_180sec].mp3`): samples "Blizzard" from Hotline Miami 2. Why: Hotline Miami is an action crime game, and as the game's author I love the 80s Vice City theme. If copying counts as a "crime", that world fits the game's satire very well. The neon look of the menus and the briefing comes from the same place. 147 BPM; the briefing and the character screen are timed to its beats.
- **Exam music** (`thrilling.mp3`): uses the opening of "Endo's Game" from the film Marty Supreme. Played quietly so the teacher's sounds are heard; it stops when you lose.

Sound	Source	When
Luigi "hmm"	file	the teacher is about to turn: stop copying
Metal Gear alert	file	the moment you lose
Chalk erasing	file (loop)	while he erases the board; its silence is a warning
School bell	file (cut to 1.8 s)	the exam title card
Button sounds	code (80s synth)	tick, choose, back in the menus
Typewriter, signature, stamp	code	the official notice
Paper tear	code (two pulls)	the caught scene

Sound	Source	When
Footsteps	code	the teacher walking
Heartbeat	code (music goes far away)	a razor-close escape
Talking blips	code (a pitch per logo)	Gemini and Claude talking
Explosion, fireworks	code	the AA grade show
Whiteboard marker	code (squeaky strokes)	"GOT CAUGHT!" written on the mugshot

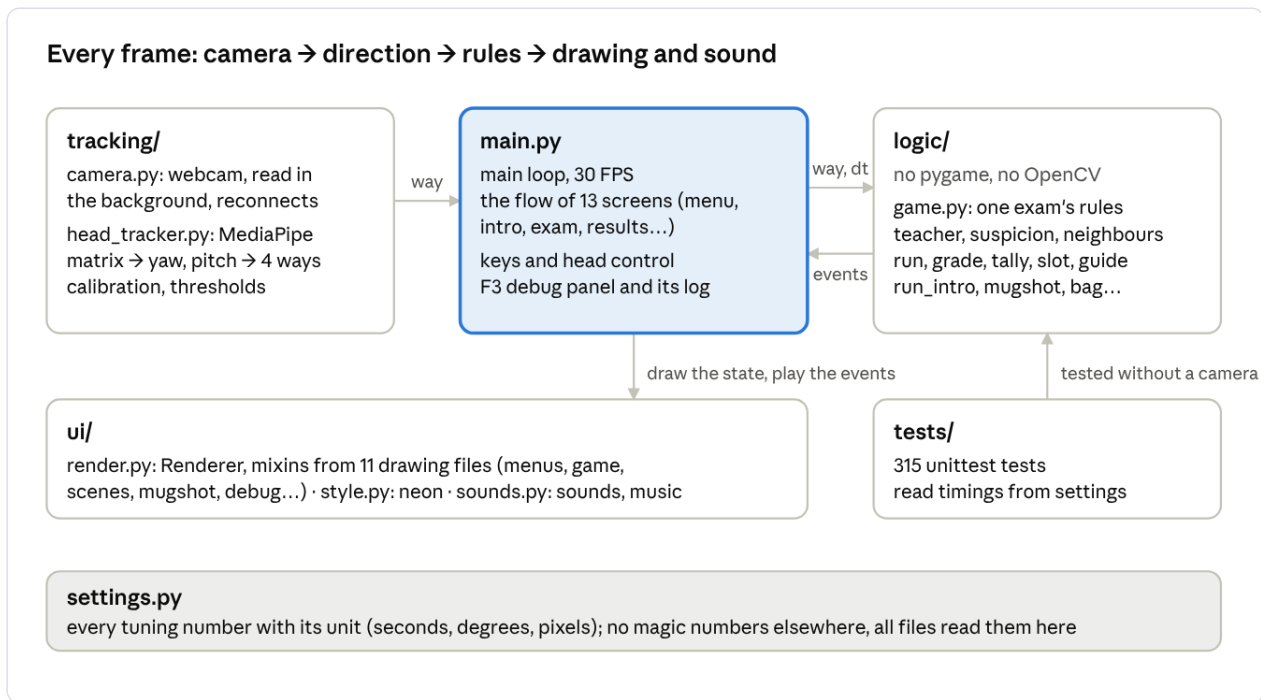
■ Interface and readability

- **In the game:** thin neon strips at the top and bottom (answers, warnings, clock; the suspicion bar), the picture in the middle. The suspicion bar turns red after 3 s; being seen copying sounds an alarm.
- **Head control made visible:** in the menus, a bar fills while the head holds a choice; a "HEAD CONTROL" / "KEYBOARD" box at the bottom right says who is in control. Every screen has key hints at the bottom.
- **Fixes for readability:** the score count went from floating points to a table (#41), gossip lines running out of their box and overlapping menu texts were fixed, long mood names are wrapped to two lines instead of shrunk.

03

Architecture today

The code is split into three folders, and one rule carries everything: the game rules (`logic/`) never import pygame or OpenCV, so they are tested without a camera or a window. This structure was set up in #25; since then every new feature fitted the same mould: numbers and timing in `logic/` (tested), drawing in `ui/`, the wiring in `main.py`, tuning numbers in `settings.py`.



Every frame, `main.py` takes the newest camera picture, `head_tracker` turns it into a direction, `game.update()` moves the rules on and returns event names (e.g. `"rip"`, `"pen"`) that are sound names too, and finally the Renderer draws the screen.

Screen flow: intro (Glitch Please) → official notice → start screen and four-pose calibration → main menu → PLAY → briefing timed to the music → character choice → for each exam: hallway gossip (slot machine) → loading and title card → exam → score table, or (after losing) mugshot and game over. After three exams: the semester's results, the letter grade and the grade roast.

Scene chain inside an exam: warning scene (the teacher comes to your desk) → on the 3rd warning or when caught: caught scene ("!" and the tearing) → mugshot → game over chat → end menu. During a scene the game is frozen; only the scene's own clock runs.

04

Summary and lessons learned

64 entries in 10 days; the busiest day was 7 October (25 entries, #15–#39), then 8 October (18 entries, update 2 and 0.1 beta). The turning points, newest first:

Date	Turning point	Entries
9 Oct	Mugshot, debug panel, epic briefing, fast-nod rule, music credits	#60–#64
9 Oct	Ten characters, the grade roast and the AA show	#58–#59
8 Oct	0.1 beta: runs on Windows without Python via <code>run.bat</code>	#55–#57
8 Oct	Update 2: four-pose calibration, characters, briefing timed to music	#40–#54
7 Oct	Hands-on guide, curved letter grades, mood pictures	#33–#39
7 Oct	Code split into <code>logic/</code> <code>tracking/</code> <code>ui/</code> ; three-exam runs	#25–#31
7 Oct	Active copying (blurry paper) and a score: the game became fun	#24
7 Oct	Neon menus, scenes, studio intro, official notice	#15–#23
6 Oct	A–D answers: read from a neighbour, write on your paper	#12–#14
2 Oct	The teacher and classroom pictures: a real game	#7
1 Oct	Head tracking (yaw/pitch → 4 directions) and the first demo	#4
30 Sep	Body-tracking demo	#1

Lessons learned:

- **Splitting rules from drawing paid off.** Keeping `logic/` free of pygame and OpenCV let 315 tests run without a camera; every new feature (slot machine, intro timing, mugshot crop) was first written and tested as plain numbers.
- **The game changed by being played.** The biggest design changes came from the team's play-tests: "copying = waiting, boring" → gradual focus (#24); "I can't write answers" → write any time, and guessing (#25); "AAAB looks like a bug" → keys without streaks (#43).
- **Not everyone sits the same.** Fixed angle thresholds fitted one player and not the next; four-pose calibration (#40), then the fast-nod rule (#63) solved it. Calibration quality is still the biggest factor.
- **Tuning numbers in one place.** All times, angles and character numbers are in `settings.py` ; balance changes (characters, moods, practice difficulty) needed no new code.
- **Making pictures is a contract with the code.** The code looks at fixed spots in the pictures, so the prompts insist that nothing moves; marking the mugshot photo's place with a pure green area removed the need to measure at all.

- **Documentation discipline.** Every change was written into `NOTES.md` the same way; this report could be written thanks to those records.

05

#1–#10

The base — from tracking to a playable game (30 Sep – 5 Oct)

In these ten entries the project went from a skeleton-tracking demo to a game with a teacher and ways to lose. Git: #1 `e99a82a`, #2 `604ed6f`, #3 `d0573bd`, #4 `d602c74`, #5 `fbbf6bd`, #6 `c401f7e`, #7 `2a61186`, #8–#10 `681fe15` (together with #11).

#1 A simple body tracker

30 September

TECHNICAL The webcam opens, MediaPipe Pose finds 33 joints and a skeleton is drawn on the live video. About 25 FPS on a 640×480 webcam.

ARCHITECTURE One file `tracker.py`, `run.sh` and `requirements.txt` for setup, the model file `pose_landmarker.task`. No game design yet; this is the tracking base.

#2 Notes and README

30 September

ARCHITECTURE `NOTES.md` (a record of every change) and `README.md`. Since then every change is recorded the same way (Summary, Added/Changed/Removed tables, Details worth knowing, Next). This report is built on those records.

#3 OpenCV package conflict

1 October

TECHNICAL `opencv-python` and `opencv-contrib-python` overwrote each other's files. Only `opencv-contrib-python` is installed now. The rule was later written into the project rules ("never add opencv-python").

#4 Game plan, head tracker and the first demo

1 October

TECHNICAL Body tracking was dropped for MediaPipe Face Landmarker. **Yaw** (left-right) and **pitch** (up-down) are worked out from the face's transformation matrix and turned into four directions: DOWN (paper), SCREEN (teacher), LEFT/RIGHT (neighbours). The first copying and staring rules, sounds and a warning box.

ARCHITECTURE The game was split into parts for the first time: `head_tracker.py` (face → direction), `camera.py`, `game.py` (rules), `sounds.py`, `render.py` (drawing), `main.py` (loop), `settings.py` (all tuning numbers) and the first `tests/`. The design went into `PLAN.md`, the steps into `STEPS.md`.

ART Everything drawn with shapes and text; no teacher yet, so you could not get caught.

#5 run.sh installs the libraries

1 October

TECHNICAL `./run.sh` sets up the virtual environment and installs missing libraries; no manual `pip install`, even after a `git pull` that adds a library.

#6 Simplified code, LEARN.md

1 October

TECHNICAL Same game, less code: the game code went from 1,066 to 856 lines.

ARCHITECTURE The team are students, so `LEARN.md` explains how the code works. Short functions, clear names and comments saying *why* have been the rule since.

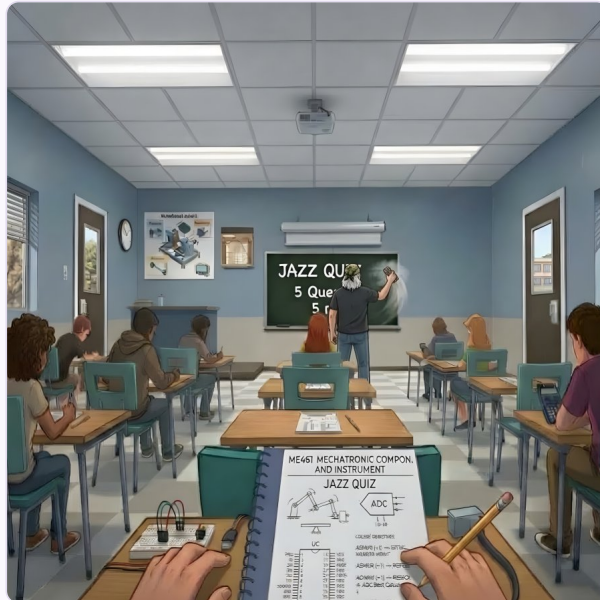
#7 The teacher, classroom pictures and sounds

2 October

TECHNICAL The demo became a real game. The teacher erases the board or plays on his phone, then turns and watches the class. Three ways to lose: **caught** copying, **3 warnings** for staring at him, the **exam clock** (60 s) running out. The classroom is only visible while the player looks at the screen; while copying the teacher can only be heard.

ARCHITECTURE `teacher.py` (the teacher's state machine: BUSY → TURNING → WATCHING) and its tests. `CLAUDE.md` holds the project rules.

ART Four classroom pictures made by the team with Gemini (teacher at the board/desk, busy/watching). Luigi's "hmm" says the teacher is about to turn; the Metal Gear alert says you are caught.



Classroom: the teacher at the board, busy (classroom_board_busy)



Classroom: the teacher at his desk, watching (classroom_desk_watching)

#8 main.py and render.py simplified

3 October

ARCHITECTURE No change in play or look; the few places where the same thing was written twice were removed.

#9 Disclaimer screen

3 October

TECHNICAL The game opens with a satirical **WARNING** screen: it is fiction, any resemblance to the professor is coincidental, we respect academic honesty. Shown once per launch. (Became the official notice in #22.)

#10 Maximized window and F11 fullscreen

5 October

TECHNICAL The game is always drawn at 960×600 inside; pygame stretches it to the window without squashing (black bars). The window opens maximized; F11 switches between borderless fullscreen and the window.

06

#11–#20

Copying, menus and scenes (5 – 7 Oct)

In this part copying became a real mechanic (A–D answers), and the game got its neon menus and first cinematic scenes. Git: #11 [681fe15](#), #12–#14 [3aaaae00](#), #15–#20 [9db2613](#) (together with #21–#23).

#11 Tuning and two test fixes

5 October

TECHNICAL New numbers in `settings.py` after play-testing. Two tests broke: one had a time written into it, the other revealed a rounding bug that stopped the suspicion bar just short of full. All 62 tests passed.

ARCHITECTURE

Tests started reading timings from `settings.py`, so tuning a number no longer breaks them.

#12 Pictures for looking down, left and right

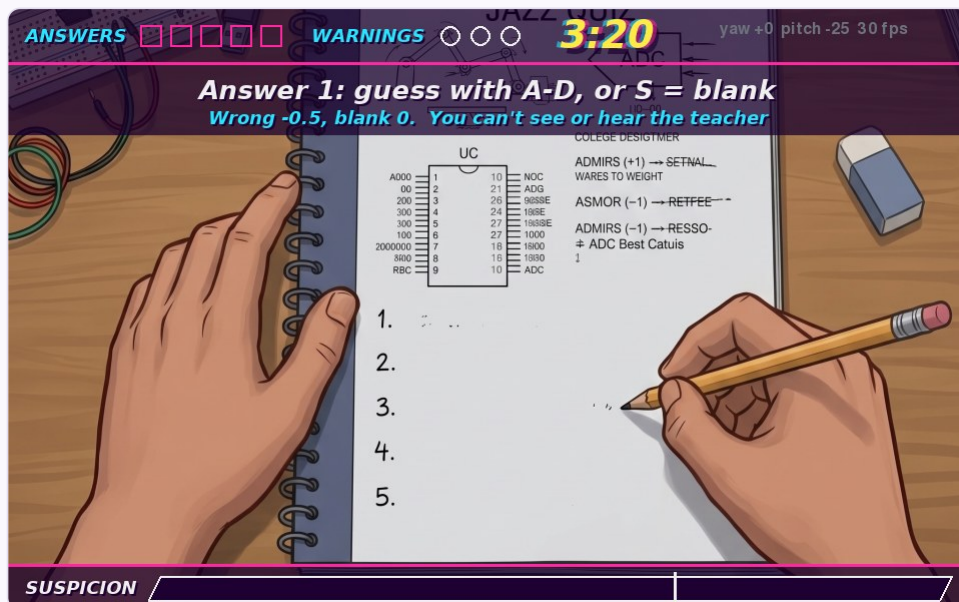
6 October

TECHNICAL

At the instructor's request, looking away shows a picture instead of a black screen: your own exam paper when looking down, that neighbour when looking left or right. None show the teacher; you still have to look at the screen (or listen).

ART

Three new pictures (`classroom_desk_looking_down/left/right`) in the same style as the classroom.



Looking down: your own paper (in-game screenshot)

#13 A-D answers: read from a neighbour, write on your paper

6 October

TECHNICAL

Copying no longer fills an answer by itself. Each question has a right letter, and only one neighbour, picked at random, knows it. Looking at a neighbour **reads** their paper (a letter or "?"); you remember the letter, look down and **write** it with A-D. After 5 answers the exam is handed in and graded (e.g. 4/5).

ARCHITECTURE

The read/write split lives in `game.py`; it closed open question Q1 in the plan.

#14 The answer on the neighbour's paper

6 October

ART The white note with the letter was replaced by pictures where the letter (A–D) or a "?" is circled on the neighbour's sheet: five per neighbour (`left_A` ... `left_D` , `left_unknown` , the same on the right). Until the paper is read the plain picture shows, so the answer cannot be seen early.



Left neighbour, A circled (`left_A`)



Right neighbour does not know: "?" (`right_unknown`)

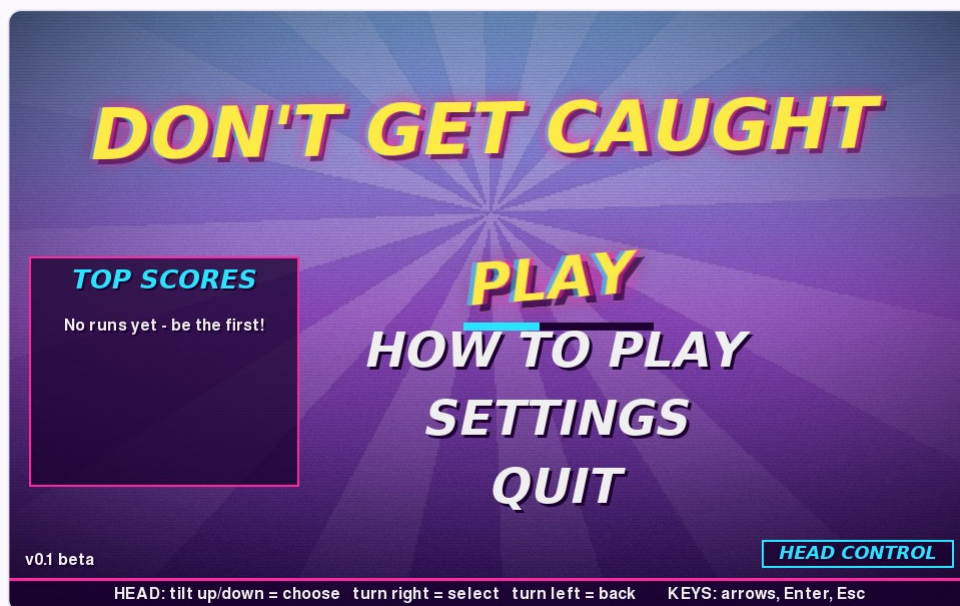
#15 A neon main menu controlled by the head

7 October

TECHNICAL Neon menus in the style of Hotline Miami: main menu (Play, How to play, Settings, Quit), settings and an end menu. Head control: tilt up/down moves the selection, turn right and hold chooses, turn left and hold goes back. Arrow keys, Enter, Esc and the mouse work too.

ARCHITECTURE `menu.py` : the menu logic (selection, hold-to-choose timing) without pygame, tested.

ART 80s neon: sunset colours, glowing text, a rocking title.



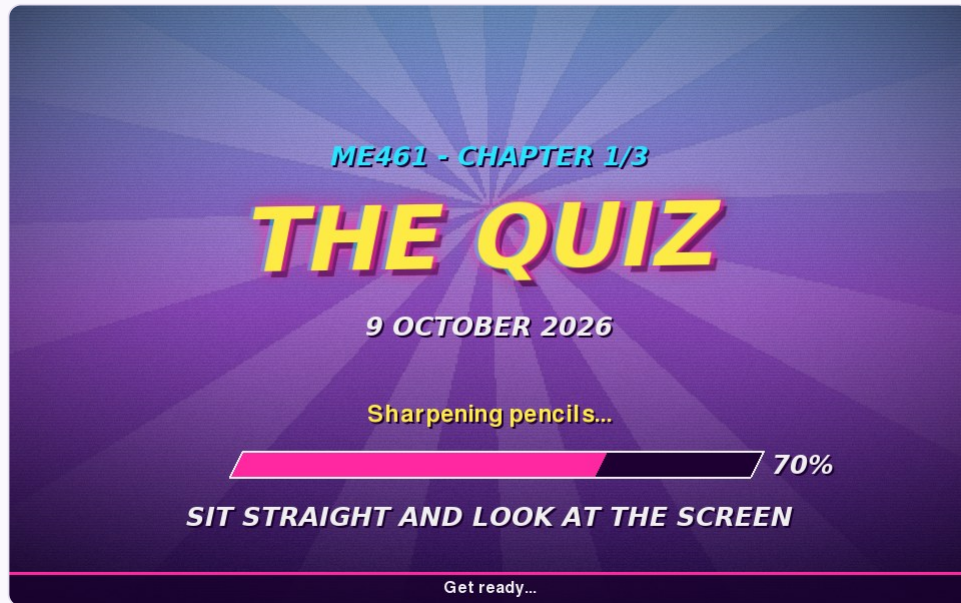
Main menu (as it is today)

#16 Loading screen and a neon game interface

7 October

TECHNICAL After Play, a 3-second "chapter" screen ("ME461 - CHAPTER 1", the exam's name, the date, a funny loading line, a bar).

ART The disclaimer, the start screen and the in-game strips, bars, warning box, pause and end screens moved to the same neon style.



Loading screen

#17 After a warning the teacher comes to your desk

7 October

TECHNICAL A warning is a scene, not a red box: the screen zooms into the teacher's picture faster and faster (as if he walks at you), then he stands at your desk pointing at you; the screen shakes and flashes red. The game freezes for 2.5 s.

ARCHITECTURE `game.py` got the idea of a **scene**: a short moment where the game is frozen and something is shown (`WARNING_SCENE`). Every later scene was built the same way.

ART The team's `classroom_warning` picture, made with Gemini.



Warning scene

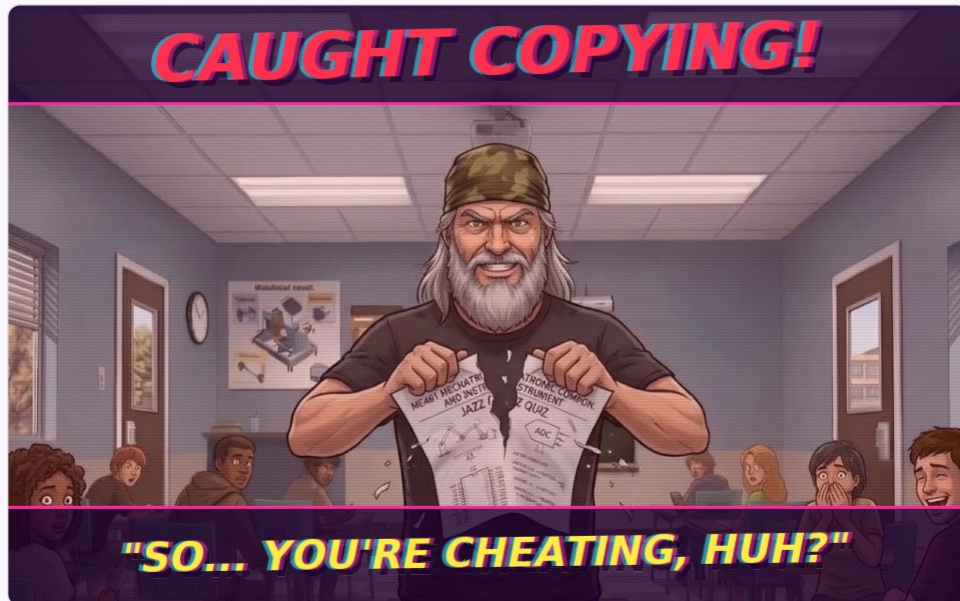
#18 Caught scene and a game over with two talking logos

7 October

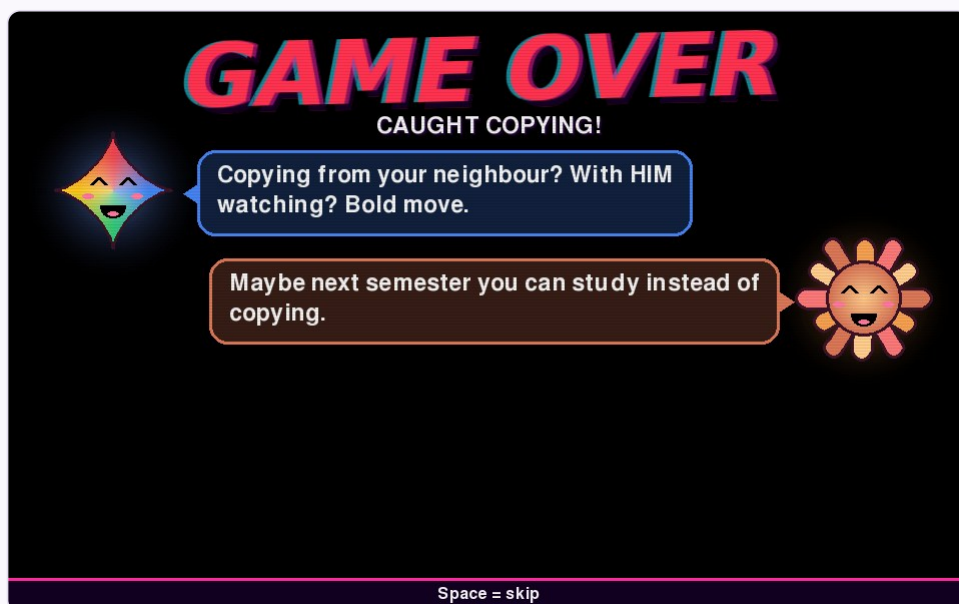
TECHNICAL When caught, the teacher freezes with a red Metal Gear "!" over his head, then a white flash, a ripping sound and he tears up your exam. After every loss a GAME OVER screen: on black, the Gemini and Claude logos make fun of you depending on how you lost. Space skips it.

ARCHITECTURE `CAUGHT_SCENE` and `GAME_OVER_SCENE` joined the scene chain.

ART The logos are drawn in code with cartoon faces. The team wanted the two AIs that made the game (Gemini the pictures, Claude the code) to laugh at the cheater.



Caught scene



Game over: Gemini and Claude talk

#19 A loading bar that fills like a real one

7 October

TECHNICAL The loading bar gets stuck, jumps, gets stuck again and makes the last jump to 100 % at the end. The "hahaha" in the game over chat is gone; both logos laugh after the second line instead.

#20 The tearing picture shows, a line per warning

7 October

TECHNICAL The tearing picture was saved as `.png` but the game only looked for `.jpeg`; it now looks for all three endings. Each warning has its own line. In the menus a key press turns head control off for 1 s, so head and keys do not fight.

07

#21–#30

Identity, score and the folder architecture (7 Oct)

In a single day (7 October) the game found its identity (studio intro, official notice), answered "it's not fun" with a score and active copying, and the code was split into today's folders. Git: #21–#23 `9db2613`, #24 `9c91a49`, #25–#30 `ed59cee`.

#21 The "Glitch Please" intro

7 October

TECHNICAL Like a studio logo before a film: on black, "GLITCH PLEASE" glitches in (red and cyan copies sliding apart, slices jumping sideways, TV static), settles, the "GP" mark above, "presents" below. About 3.4 s, with a sound made in code. Any key skips it.

ARCHITECTURE One file, `glitch_intro.py`, needing only pygame, so it can be copied into all the team's projects.

#22 The disclaimer becomes an official notice

7 October

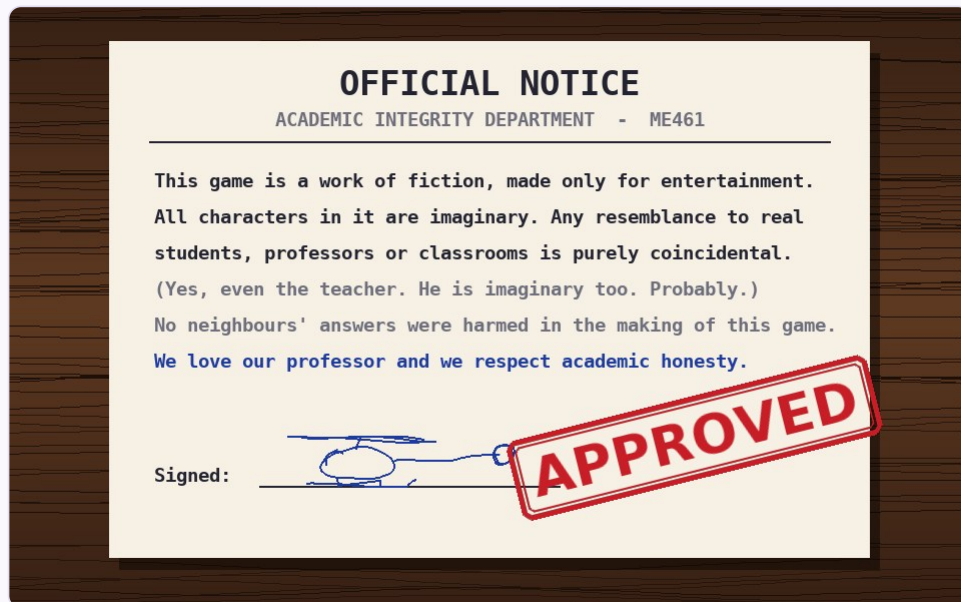
TECHNICAL A cream paper slides onto a wooden desk: "OFFICIAL NOTICE — ACADEMIC INTEGRITY DEPARTMENT - ME461". Six lines are typed out in a typewriter font with key sounds. Space signs it (a blue scribble; really someone trying to draw a helicopter), then a red "APPROVED" stamp slams down with a thump and the paper shakes.

ARCHITECTURE

`disclaimer.py` keeps the timing and returns events ("type", "sign", "stamp"); the drawing lives in another file. The rules/drawing split makes it testable.

ART

Deliberately different from the neon menus: paper, ink, stamp. Even the wooden desk is drawn in code, row by row.



Official notice: signed and stamped

#23 Softer, smoother menus

7 October

ART

The Hotline Miami look stayed, with many more tones: a three-colour sunset gradient drawn pixel row by pixel row, soft-edged light rays, a light film grain hiding colour steps, fainter scanlines, darker corners, a blurred glow behind the text. Big screens now scale smoothly.

#24 Gradual focus, camera reconnecting, a score

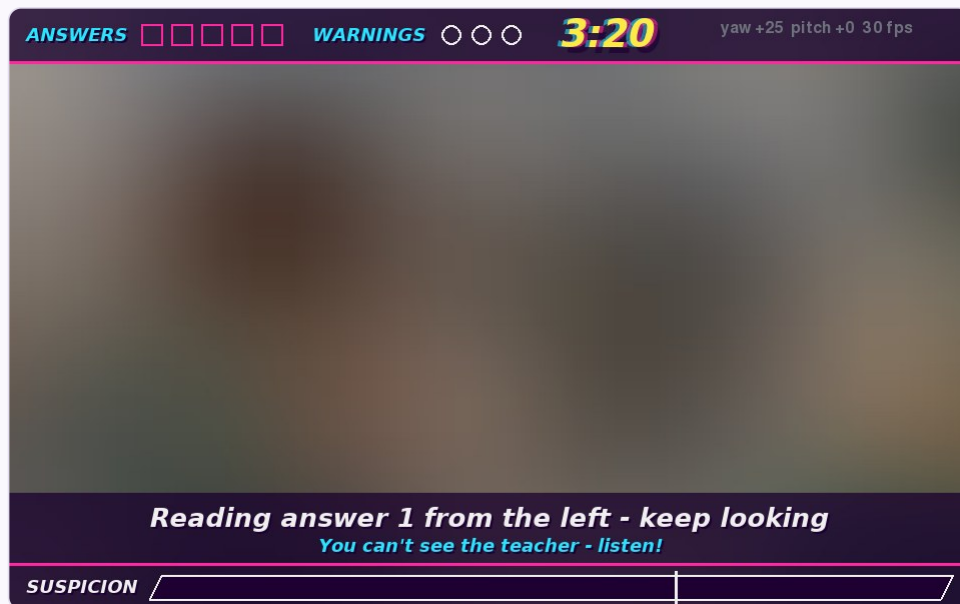
7 October

TECHNICAL

The team found the game not fun (copying = waiting). Two fixes: **copying became active**, the neighbour's paper starts blurry and sharpens over 2.5 s of looking; every look starts blurry again, so reading means holding a risky look. **A score** arrived: right answers, time left, getting away after being seen copying (+150), warnings; the best score is saved. When the camera drops out the game no longer closes; it reconnects by itself.

ARCHITECTURE

Gradual focus and the camera code came from teammate Emre's branch; it had gone in another direction, so it was not merged with git: the ideas and the camera code were carried over by hand. `highscore.py` added.



Looking left: the neighbour's paper is still blurry

#25 Guessing, blanks, Balatro-style counting, folders

7 October

TECHNICAL

A player could not write answers after the first question; the likely cause was the rule "write only after reading the neighbour who knows". The rule went: **any letter can be written at any time**, and a question can be left **blank** (S). Grading: right **+1**, wrong **-0.5**, blank **0**; a blind guess is a gamble. When time runs out the paper is **collected** and graded (no longer a loss). The score is counted up part by part, like Balatro.

ARCHITECTURE

Today's structure was born here. `game.py` (323 lines) and `render.py` (1,456 lines) were split into small single-job files and sorted into three folders: `logic/` (rules, no drawing: `exam_paper`, `neighbours`, `suspicion`, `tally`, `game`), `tracking/` (camera, head), `ui/` (`style`, `draw_game`, `draw_menus`, `sounds`). The rule: `logic/` never imports pygame or OpenCV, so it is testable without a camera or a window.

#26 Three exams per run, early bonus, top scores

7 October

TECHNICAL A run is now **three exams**: the Quiz (3 questions, 150 s), the Midterm (4, 130 s), the Final (5, 120 s). Each exam the teacher has a **mood** (busy/watching times, how often he moves), told on the loading screen as "hallway gossip". A failed exam scores 0 but the run goes on. After the Final the total is counted up next to the **top 5 scores**. The escape bonus grows with how full the suspicion bar was (100–500; above 80 % "RAZOR CLOSE!" +300).

ARCHITECTURE `logic/run.py` (the three-exam run), `ui/draw_results.py`.

#27 The hallway gossip gets its own screen, button sounds

7 October

TECHNICAL Before each exam a gossip screen: a short story of what happened to the teacher and **what it means** for the exam, then "I'M READY".

ART 80s synth sounds for the buttons: a tick when the selection moves, three notes up when choosing, two down for back. All made in code.

#28 Background music

7 October

ART The team made an 80s Miami theme with Suno by sampling "Bastır" by Murda and EGE! (`theme.mp3` , 190 s). It loops after the notice is signed; not during the intro and the notice, which have their own sounds.

#29 Music only in the menus, smooth screen changes

7 October

TECHNICAL The music fades out over 1.5 s on the loading screen, the exam is silent, it comes back on the results. Every screen change is a 0.4 s crossfade. The game over chat has 8 conversations per way of losing instead of 1.

ARCHITECTURE `logic/bag.py` : a "bag" — every joke comes once before any repeats, never the same twice in a row. Later used in many places.

#30 A slot machine for the mood, good and bad days, the hidden suspicion line, bonuses

7 October

TECHNICAL The mood is drawn on the gossip screen by a **slot machine** ("Spin" with the head), so the player sees it is random. 13 moods in all: the Quiz's are good days, the Midterm's have good and bad sides, the Final's are bad days. **He keeps an eye on you:** above a hidden point on the suspicion bar (60 % / 45 % / 30 %) the teacher keeps watching until the bar drains below it. Bonuses: **NINJA!** +1500 (all right, no warning), **ALMOST NINJA** +500, **SHARP EYE** +100.

ARCHITECTURE `logic/slot.py` (the reel's physics), `ui/draw_briefing.py`.

08

#31 – #40

Polish, the guide, grading and personal calibration (7 – 8 Oct)

In this part the game learned to explain itself (a hands-on guide), to grade the player (a curve) and to fit each player's way of sitting (four-pose calibration). Git: #31 `ed59cee`, #32 `96f48a3` + `6971bcf` (merges), #33–#39 `f79da81`, #40 `b6a7d33` (update 2).

#31 The slot machine looks like a slot machine

7 October

ART "MOOD-O-MATIC": a gold cabinet with bulbs that blink while waiting, chase while spinning and flash when it stops; a paper-white drum darker at the top and bottom, text blurring when spinning fast; a red-ball lever that the head pulls down. The reel goes a little too far and settles back, like a real one.

ARCHITECTURE The overshoot and settle are in `logic/slot.py`, tested.

#32 The Windows branches merged into main

7 October

ARCHITECTURE `windows` (the Windows launcher `run.bat`) and `windows-support` (DirectShow camera, loading the model as bytes) were merged into `main` with real merges. Both started before the code moved into folders, so their changes were carried over to the new structure by hand. Since then the game is developed on `main`.

TECHNICAL MediaPipe is given the model file's bytes instead of its name; otherwise it could not load on Windows computers with `ç`, `ş` or `ı` in the folder path.

#33 Exam music, talking blips at game over

7 October

ART The team's `thrilling.mp3` (made with Suno from the opening of "Endo's Game", from the film *Marty Supreme*) plays during an exam, quieter so the teacher's sounds are heard, and stops when you lose. The game over bubbles make small talking blips while typed, each logo in its own voice. The Metal Gear joke was removed at the team's request.

#34 A hands-on guide with Gemini and Claude

7 October

TECHNICAL "How to play" was nine lines of text. Now Gemini and Claude take turns talking and finishing each other's sentences while the player tries each thing with their head. Behind them is what the player would see in the game. 12 steps cover the whole game.

ARCHITECTURE `logic/guide.py` (steps, tasks, typing timing; tested) and `ui/draw_guide.py`.

#35 After the guide, back to the menu (crash fix)

7 October

TECHNICAL The game closed when the guide ended: on the last frame the drawing asked for a step that no longer existed (`IndexError`). Now the guide goes straight back to the main menu.

#36 Three-letter arcade names

7 October

TECHNICAL A run that makes the top scores asks for a three-letter name, like an arcade cabinet. With the head: up/down rolls the letter, right goes to the next, left goes back; typing works too. `q` and the window's X no longer quit.

ARCHITECTURE `logic/name_entry.py` (tested).

#37 The semester's letter grade, on a curve like a real teacher

7 October

TECHNICAL The run ends with a METU-style letter grade (AA...FF), stamped after the count. It is given **on a curve over all earlier runs on this computer**: how many standard deviations the run's total is above their average (z). An average run is a CC. For the first 5 runs there is no "class" yet, so a fixed table is used (90 % = AA, 50 % = CC).

ARCHITECTURE `logic/grade.py` (tested).

#38 Chalk, footsteps, a real tear; mood pictures

7 October

ART The board-erasing sound loops while the teacher erases and stops the moment he starts to turn, so silence is a warning too. Footsteps when he moves and when he walks to your desk after a warning. The tear is two pulls (rrrip... rrrrip!). Footsteps and the tear are made in code until files arrive.

ARCHITECTURE Mood-specific pictures: if the file exists, `<picture>_<mood>` is shown (a party hat on his birthday, a suit for the dean's visit...).

#39 Gemini prompts for the mood pictures

7 October

ARCHITECTURE `IMAGE_PROMPTS.md` : for each picture, which image to upload, a ready-to-paste prompt and the file name. 16 prompts.

ART

The prompts insist that nothing in the scene moves: the warning scene zooms into the teacher's face at fixed spots, the neighbours' papers are looked for at fixed places.

#40 Four-pose calibration

8 October

TECHNICAL

Calibration only measured "looking at the screen"; everyone had to turn 18° to copy and tilt 23° to look down. Now four poses are measured in turn: screen, left, right, down, each averaged over 1 second. The thresholds are 60 % of each pose, between 8° and 35°. The game fits how the player sits.

ARCHITECTURE

`Calibration` and `pose_threshold()` in `tracking/head_tracker.py` (tested).

09

#41 – #50

Update 2 — characters and a briefing timed to music (8 Oct)

Update 2 made the game more replayable: a readable score table, a practice exam, a score that spins like a slot machine, characters that bend the rules and a briefing that sits on the music's beats. Git: #41–#50 are all in [b6a7d33](#).

#41 The score count as a table

8 October

TECHNICAL

The count after an exam was question cards with points floating over them; the team found it hard to read. Now it is two tables, still counted up like Balatro: on the left a row per question (Q1, what you wrote, the answer key, RIGHT / WRONG / BLANK, points) with the grade under them; on the right a row per bonus. The row whose turn it is glows in its colour; the answer key stays "?" until its turn.

#42 A practice exam after How to play

8 October

TECHNICAL The guide now leads into a two-question, 60-second **practice exam**. The teacher is in a new "practice" mood (reading with his tea, never leaving the board). A normal exam in every way, but it counts for nothing: no top score, no class average, no grade.

#43 Answer keys without streaks

8 October

TECHNICAL Drawing each letter on its own made "A A A B" common, and the team thought it was a bug. The key is still random but never repeats a letter twice in a row, and no letter appears more than twice in an exam. The neighbour who knows is never the same side three times in a row.

#44 The slot-machine score

8 October

ART The big score is a reel per digit: while counting they all spin (units fastest, blurred), and when they stop each reel clunks onto its digit. The show grows with the score: the reels shake harder, the bulbs chase faster, more sparks fly. A huge score ends with "JACKPOT!".

#45 Characters

8 October

TECHNICAL Before a run the player picks who they are on a "WHO ARE YOU?" screen. Each character bends one rule with an advantage and a price. Examples: **Glasses** reads 25 % faster, but every look at the teacher blurs the classroom for 0.9 s. **The nerd** has one joker per exam but loses 1000 points if he does not hand in with 30 % of the time left. **The energy drink addict** tosses a coin each exam: a sugar rush (the world at 85 %) or a crash (sleepy spells, reading at 40 %).

ARCHITECTURE Every number is in one table, `CHARACTERS` in `settings.py` ; the rules read it through `logic/character.py` , so adding a character needs no new code.

ART The portraits are not pictures but neon heads drawn in code (cap, glasses, can...).

#46 The New Era guy and the Monster laptop

8 October

TECHNICAL The cap's creeping bar fills in 30 s instead of 45; in exchange his normal bar fills slower.

ART "The guy with the cap" is now **the New Era guy**: flat brim, gold sticker still on. The "Monster bag" is not an energy drink but a bag for a **Monster gaming laptop** (its edge glows in changing colours). Every character line was rewritten to be funnier.



Character choice: the New Era guy

#47 Fewer sugar rushes, an easier practice, no face on screen

8 October

TECHNICAL The chance of a sugar rush drops 15 % after each rush (50 % → 35 % → 20 %). The practice exam runs at 75 % difficulty. The guide got the hidden-suspicion-line joke ("I put a random line in the code." "Where is it?" "No idea. I forgot. Good luck!").

ART The webcam picture was removed from the game, the guide and the menus; it only shows while calibrating.

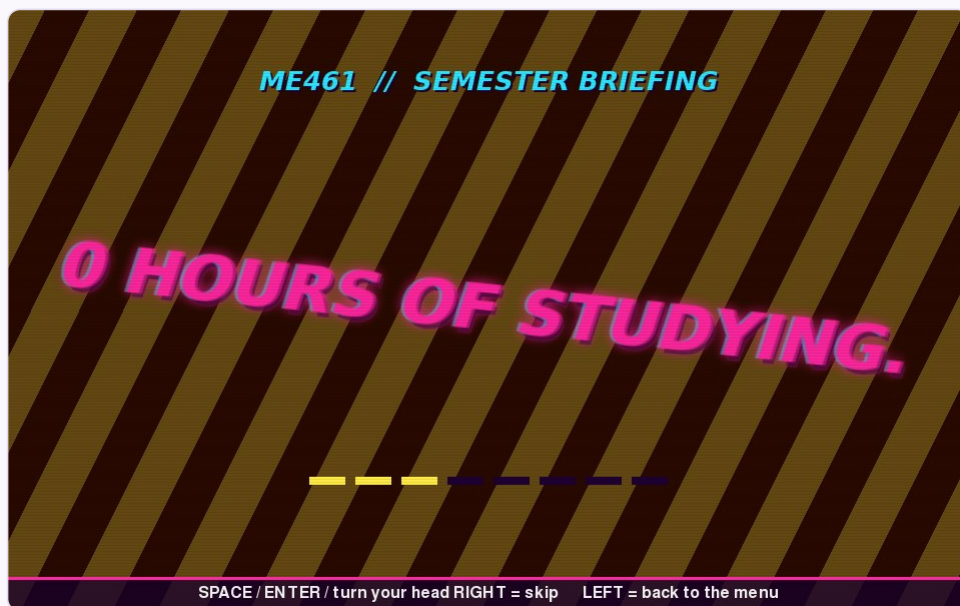
#48 A briefing timed to the character music; the sliding character screen

8 October

TECHNICAL The team added a track for choosing the character (made with Suno by sampling "Blizzard" from Hotline Miami 2). Claude cannot hear, so the track was **measured** with numpy: the autocorrelation of the sound's onsets gave a tempo of **147 BPM**; the strongest hits come every 1.633 s (a bar of 4 beats) from **0.81 s**; the loudness doubles at **11.84 s** (the drop), on the same beat grid.

ARCHITECTURE A new screen `RUN_INTRO` and `logic/run_intro.py` for the timing; drawing in `ui/draw_run_intro.py`, the character screen in `ui/draw_characters.py`.

ART On every bar's hit a new line slams onto the screen, huge and tilted, with a white flash and a shake, over diagonal stripes whose colours change every line: "3 EXAMS." "1 SEMESTER." "0 HOURS OF STUDYING..." At the drop the character row slides in from the right; the chosen one is big, in a yellow frame, thumping on every beat.



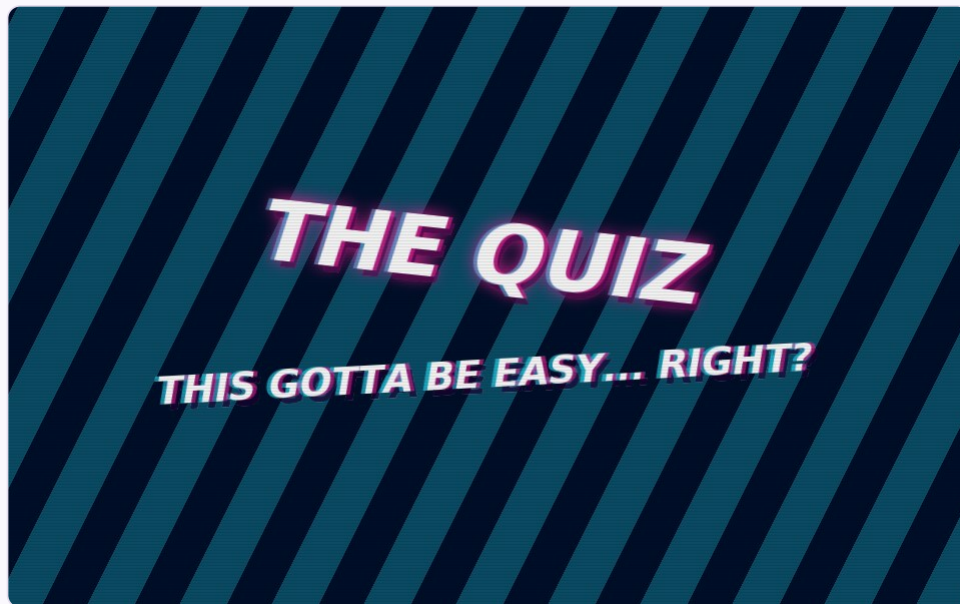
Briefing: one line per bar

#49 Exam title cards

8 October

ART When the loading bar fills, a 1.8 s card: the exam's name slams in with a flash and a thud, then its ironic line: "THE QUIZ — This gotta be easy... right?", "THE

MIDTERM — I can handle this. Probably. Maybe not.", "THE FINAL — God, please help me." The first exam's gossip keeps the character music going. The briefing now shows only the newest line.



Title card: THE QUIZ

#50 The school bell, a sideways character row, "Are you sure?"

8 October

TECHNICAL The team's school bell file (3.4 s) rings only on the title card, cut to the card's length (1.8 s) and faded out, so bell and card end together. The character row is sideways: turn left/right to browse, look down and hold to choose. An "ARE YOU SURE?" box for QUIT and MAIN MENU says that leaving mid-run loses the run.

10

#51 – #59

The slot machine, version 0.1 beta and the semester show (8 – 9 Oct)

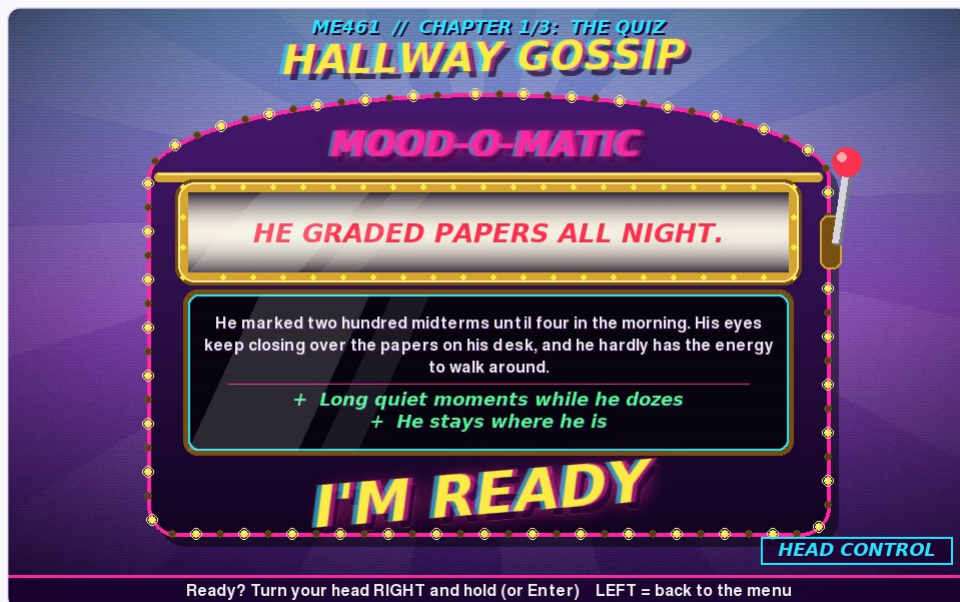
In this part the gossip screen became a real gambling machine, the game learned to run on Windows computers without Python, and the semester grade became a

show. Git: #51–#54 [b6a7d33](#) , #55 [0607354](#) , #56 [273aad2](#) , #57 [e95a6c7](#) (all three on the [release-0.1-beta](#) branch); #58–#59 are not committed yet.

#51 A real slot machine for the gossip; the helicopter signature

8 October

ART The gossip screen became one slot machine, drawn entirely in code: a domed cabinet, a neon edge, a shadow, "MOOD-O-MATIC" glowing on the dome, a gold-framed reel with a glass gleam, the lever at its side. Under the reel a screen built into the machine: before the spin it blinks "PULL THE LEVER", after it the story and the +/- lines light up there. The notice's signature is now a shaky hand trying to draw a helicopter, stroke by stroke.



Hallway gossip: the MOOD-O-MATIC

#52 Fix: the machine's screen stayed dark

8 October

TECHNICAL When the reel stopped, its clock stopped too; but the screen lights up after the stop, so it stayed dark and the bulbs flashed forever. The clock now keeps running after the stop.

#53 Casino bulbs round the slot machine

8 October

ART A small yellow bulb every 24 pixels along the neon edge; every other one lit, swapping every 0.35 s (three times as fast while spinning), so the light seems to run along the edge.

#54 The moods do what their gossip says

8 October

TECHNICAL The team asked whether the teacher really behaves as each mood's +/- lines promise. Every line was checked against the numbers (`normal_day` as the reference). The ones that did not match were fixed: e.g. moods promising "long, angry looks" now look 18 % longer instead of 5 %. For the Final's moods the line was removed rather than the looks made longer; the game is hard already.

#55 0.1 beta as a Windows .exe

8 October

ARCHITECTURE One .exe with everything inside, for computers without Python: GitHub built it on its own Windows machines on every push, running the tests first. At about 200 MB (over GitHub's 100 MB limit) the .exe was never committed. (Dropped in #56.)

#56 No .exe: run.bat installs Python by itself; a sturdier Windows camera

8 October

TECHNICAL `run.bat` looks for a usable Python (an existing `.venv`, then `py -3.14/3.13/3.12`, then `python`; 64-bit, 3.10–3.14). If there is none it installs **Python 3.14 for this user only, without an admin password** (winget, or the python.org installer via PowerShell). Windows has two camera systems: DirectShow (fast to open) and Media Foundation (slow but works with every webcam). Each webcam is now tried with both; 640×480 and a one-frame buffer are requested (less delay).

ARCHITECTURE The .exe build code was removed; the version name "0.1 beta" stayed.

#57 **run.bat stops with a clear message when it is not next to the game**

8 October

TECHNICAL First try on a Windows without Python: Python installed, then "requirements.txt not found". Most likely `run.bat` was double-clicked inside the ZIP (Windows then copies only it to a temporary folder). Now `run.bat` first checks that `main.py` and `requirements.txt` are next to it; if not, it installs nothing and says to unzip first.

#58 **New characters, a short ironic guide, a heartbeat**

9 October

TECHNICAL Ten characters in all. New ones: **the 7th-year legend** (older than the professor; the teacher's looks are 25 % shorter, but the bar fills $\times 1.3$ when seen), **not a ME student** (took the course as an elective; the teacher thinks he is lost, so staring fills the bar at $\times 0.45$, but the neighbours' answers are Greek to him: $\alpha \beta \gamma \delta$), **the "quick question" guy** (hand always up; the teacher is busy 30 % longer but the bar creeps up while he is not looking at his paper), **the front-row student**. A heartbeat now plays on razor-close escapes and the music turns muffled and far away. The guide went from 41 lines (about 2 min) to 23 (about 45 s).

ART The characters are types from a real class; none is named with an insult or could point at real classmates (some ideas were dropped on purpose). The notice was rewritten: all characters are imaginary ("Yes, even the teacher. He is imaginary too. Probably."). 8 new pictures for the Greek letters.



For the not-a-ME-student: β on the neighbour's paper (left_greek_B)

#59 The grade roast after the Final, a show for good grades

9 October

TECHNICAL When the semester grade is stamped after the Final, Gemini and Claude come back and roast the letter in two or three ironic lines ("CC. Wow, I thought you were special." / "NAAHHH. Average. Always was."). It can be turned off in Settings.

ART A show that grows step by step for good grades: **BB** the letter slams down big, "RESPECTABLE.", a little confetti; **BA** bigger, pink to yellow, turning light rays, a fanfare; **AA** an explosion and a white flash, the screen shakes, gold letters, fireworks everywhere (each with its bang and crackle), "LEGENDARY CHEATER".

11

#60 – #64

This session — the epic briefing, the mugshot, the debug panel (9 Oct)

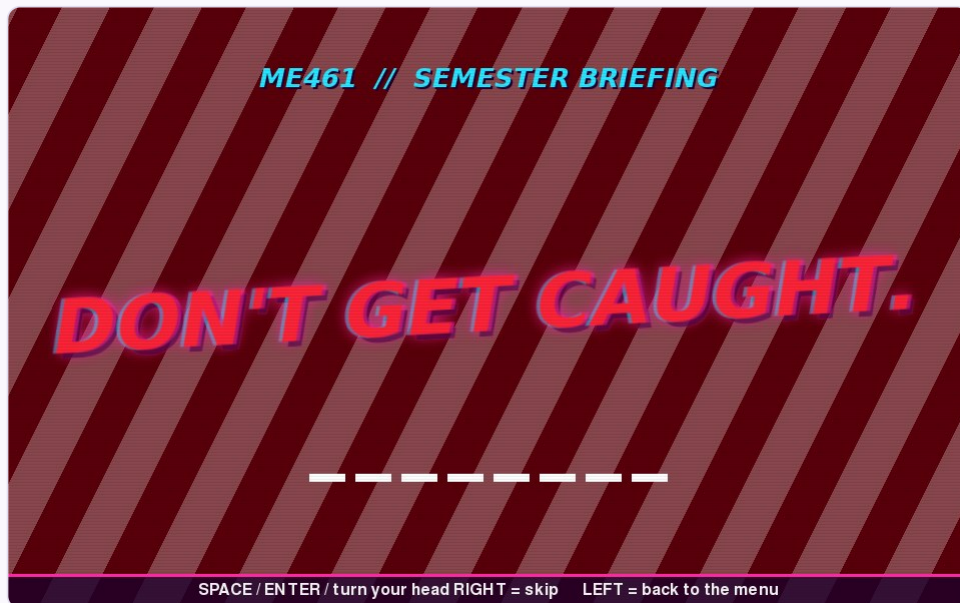
All five changes were made on 9 October and are not committed yet; the last (#64) is documentation only. Two face the player (briefing, mugshot), two help explain and debug the project (debug panel, fast-nod rule). All 315 tests pass.

#60 The briefing goes from soft to hard and ends on "DON'T GET CAUGHT."

TECHNICAL The lines still come one per bar, evenly (a first try sped the last lines up; the team did not want that, only harder hits). The first line lands softly; each line hits harder than the one before: size, shake, flash, beat thump and stripe speed rise linearly from the first line's value to the last's (e.g. shake $\times 0.3 \rightarrow \times 2.5$). A new last line, "**DON'T GET CAUGHT.**", comes on the next bar, one beat after the drop (12.24 s), and stays 8 beats instead of 4.

ARCHITECTURE On the rules side `intensity(line)` gives a number rising from 0 to 1 (`logic/run_intro.py` , tested); on the drawing side each effect is worked out with `grow()` from a (soft, hard) pair. How long the title stays is `INTRO_TITLE_BEATS` in `settings.py` .

ART The title fills the width, swaps red and white on every beat and rocks left and right.



The briefing's last line

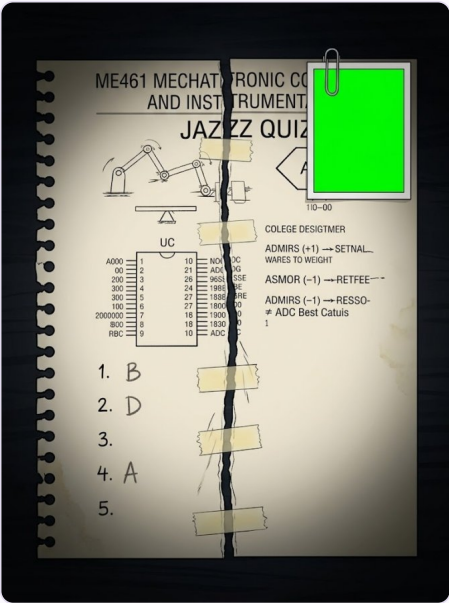
#61 The mugshot: your photo on the torn exam

TECHNICAL The moment you lose (caught, or the 3rd warning), the camera takes a frame. The photo is cropped around the face: the box around MediaPipe's 478 face points becomes a cut-out 1.9 times the face's height in the photo's shape, always inside the picture. On a black, silent screen the paper fades in over 1.5 s; 2.5 s later "GOT CAUGHT!" is written from left to right with the squeak of a whiteboard marker. The photo is only kept in memory, never written to disk.

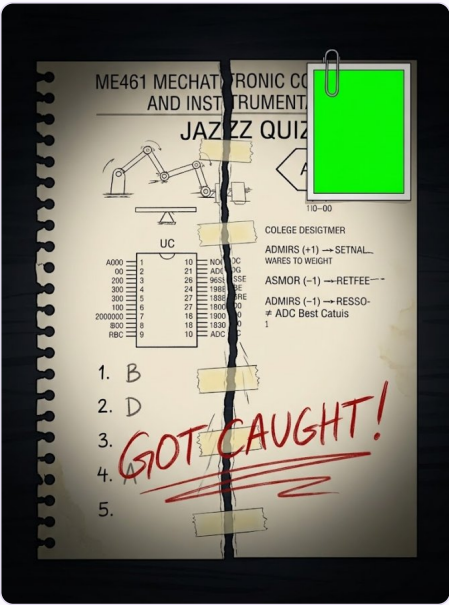
ARCHITECTURE A new scene `MUGSHOT_SCENE` between the caught scene and game over. `logic/mugshot.py` does the crop (tested), `ui/draw_mugshot.py` the drawing. The photo's place is found by the colour of a **pure green (#00FF00) rectangle** in the picture; the photo is put only on the green pixels, so the paperclip stays on top. A new picture needs no measuring. The "writing" effect is the written picture revealed over the plain one, left to right, only between the columns where the two differ. The marker sound is made in code: 12 strokes of random length, the felt tip's hiss (high-frequency noise) and a squeak with a wobbling pitch.

ART The paper is two pictures the team made with Gemini: the game's JAZZ QUIZ paper, torn vertically and taped, under a single interrogation lamp in a dark room; in the second the teacher has written "GOT CAUGHT!" in red. The prompt took a few

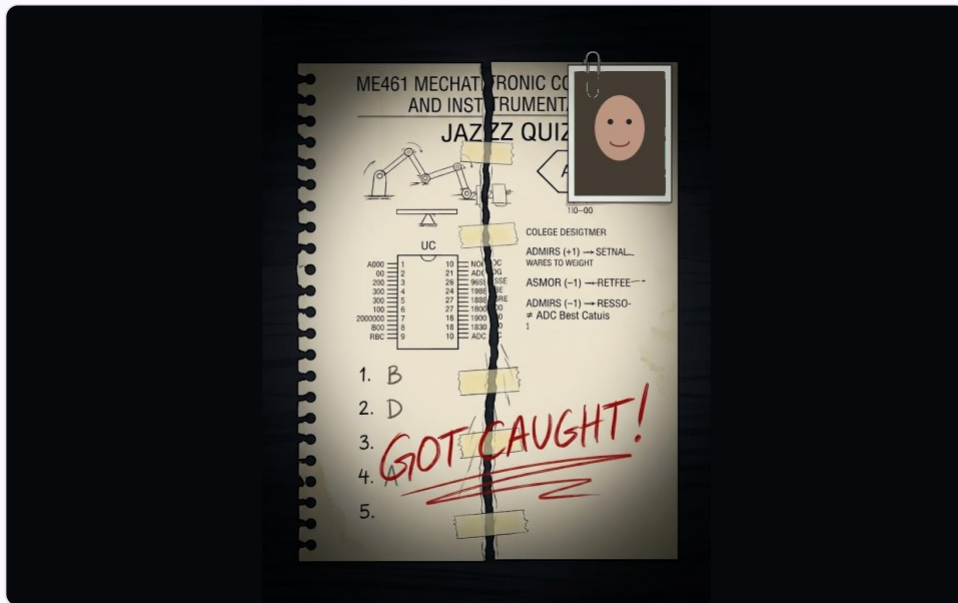
rounds: first realistic with a horizontal tear, then pulled to the game's comic style, a vertical tear and a dark theme.



The mugshot paper (torn)



The same paper, written on (got_caught)



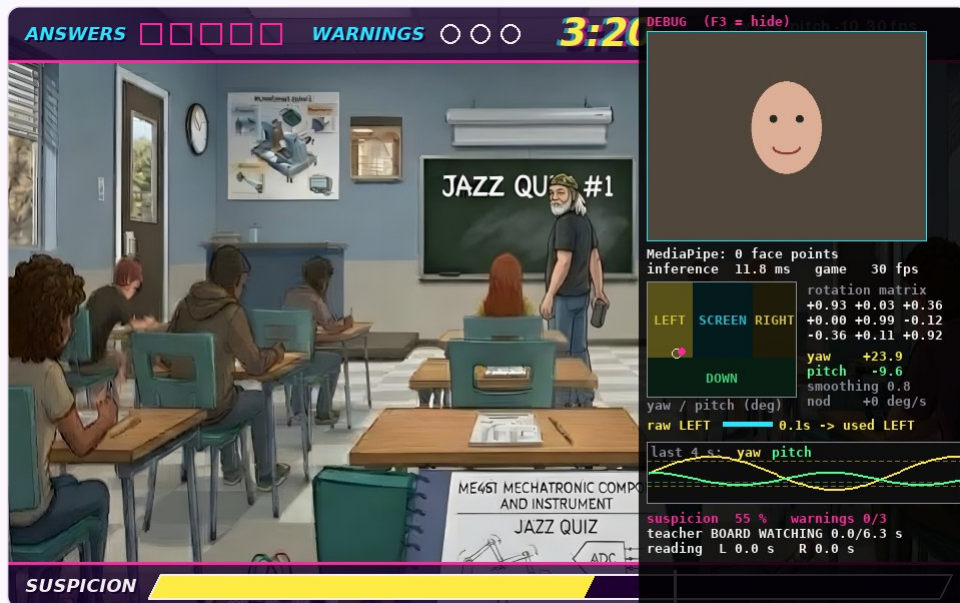
The mugshot in the game (with a sample face)

#62 The debug panel

F3

TECHNICAL F3 opens a technical panel on the right on every screen, showing the data's path each frame from top to bottom: (1) the camera picture, MediaPipe's face mesh, the face box, the nose arrow, inference time (ms) and FPS; (2) MediaPipe's 3×3 rotation matrix and the yaw/pitch worked out from it; (3) a yaw/pitch chart split by the calibrated thresholds into LEFT / SCREEN / RIGHT / DOWN, with the raw (hollow) and smoothed (full) dot; (4) the decision chain: raw direction → HOLD_TIME bar → direction used; (5) the last 4 s of yaw and pitch, thresholds dashed; (6) during an exam, suspicion, the teacher and reading times.

ARCHITECTURE `ui/draw_debug.py` ; the panel only reads and changes nothing in the game. It draws in under 1 ms. The face mesh is drawn on a copy of the camera picture, so the mugshot gets a clean photo.



The debug panel over the game screen (sample data)

#63 A fast nod down counts as **DOWN**; the panel keeps a log

TECHNICAL Sometimes looking at the paper did not count: on a quick nod MediaPipe loses the face before the head is 8° down, and the old rule ("lost while already tilted down") did not fire. With the team's idea the tracker now also looks at **which way and how fast** the head was moving: if the pitch fell faster than 40° per second over the last 0.3 s, it is **DOWN**. While F3 is on, every frame is written to `debug_log.csv` (angles, nod speed, direction, status note).

ARCHITECTURE `tracking/head_tracker.py` : `remember_pitch()` , `pitch_speed()` , `went_down()` (4 new tests). The team tried it: after recalibrating, looking down worked well; the real cause was most likely a **DOWN** pose not held far enough down.

#64 Music sources in the README

ARCHITECTURE A source table was added to the README's music section: all three tracks were made with Suno by sampling "Bastır" (Murda and EGE!), "Blizzard" (Hotline Miami 2) and "Endo's Game" (Marty Supreme); the reason for choosing Hotline Miami is written there too. No code changed.